

Siyuan Jiao

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Education

- **UC San Diego, Computer Engineering**
 - Pursuing BS in Computer Engineering, expected to graduate 2029
 - Notable courses: CSE11 Accelerated Intro to Programming, CSE12 Intro to Data Structures, Math 20D Introduction to Differential Equations, Math 11 Calculus-based Statistics
- **San Diego State University** **2024-2025**
 - Math 252 Calculus III, Math 254 Introduction to Linear Algebra

Skills

- Programming in Java, Python, C#, GDScript, Luau using Unity, Godot Engine, Roblox Studios, and Blender. In addition, familiarity with Wolfram Mathematica, Matlab, and R.
- Game design principles and experience, including art direction, programming, and GUI design
- Fundamental understanding of electrical engineering and transistor/chip design, PC building experience
- Leadership experience through serving as secretary and moderator to club of 40+ members, leadership role in orchestra, and tutoring experience.
- Proficiency in both the English and Mandarin language.

Experience

- **Student Researcher, COSMOS at UCSC, Cluster 10: Semiconductor Materials and Device Engineering** **Summer 2024**
 - Studied semiconducting devices/transistors through topics such as band diagrams, PN-junctions, and metal-semiconductor contacts, examined the fabrication process behind producing MOSFET transistors
 - Researched graphene transistors (GFETs) as a possible replacement to silicon in future semiconducting devices, investigated properties such as electron mobility, analyzed solutions to current limitations. Presented findings to multiple professors at UCSC.
- **Programming** **2022-Present**
 - Created a variety of personal projects using multiple platforms, implemented features such as 2D/3D animation, motion control, and sound design
 - Earned Coursera certificate in the UMich Python Basics course
- **Hardware Experience** **2024-Present**
 - With the C2KSD organization, refurbished devices, including laptops and PCs, for donating to 300,000+ families in need in order to address the digital divide,
- **UCSD Video Game Development Club** **2025-Present**
 - Member of VGDC, participated in game jams and development projects, created visual novel with integrated minigames, actively attended various workshops exploring the different aspects of game development, including scripting, world-building, and sound design
- **Project Butterfly** **Jan-Dec 2023**
 - Tutored English to 30 children and adults in Asia through weekly online lessons using Zoom, 50+ hours, created educational material and lectures, aided adult students in applying English skills to business platforms
 - Organized book fundraiser in partnership with the African Library Project to help create libraries in Sub-Saharan Africa

Additional Experience

- **Musical Experience** **2021-Present**
 - San Diego Youth Symphony: 2 years, played leadership role of assistant principal 2nd violin in Sinfonia division in 2022- 2023, guided section by providing instructions for fingerings/bowings, advised and demonstrated techniques for new players, performed in various concert halls across San Diego
 - 9+ years of violin experience, independently learned guitar (5 years), bass (3 years), and piano (2 years)
- **Art Experience** **2020-Present**
 - 6+ years of art experience. Proficiency with digital art programs such as Clip Studio Paint, Procreate, Adobe Fresco, and ibisPaint X as well as traditional media such as acrylic, watercolor, charcoal, and pastel. Created several award-winning pieces.
 - Created art social media content that reached 600k+ audiences